CUDA Part II — Kernels, Sorting & Bandwidth Profiling

Thread Tasks, Parallel Merge Sort & Memory Bandwidth Analysis

CUDA C (nvcc 13.2) Status

> **UCS645: Parallel & Distributed Computing | Assignment 7 — CUDA Part II**

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System Configuration

Component	Details
CPU	Intel Core i7-14700HX (20 cores, 28 threads)
GPU	NVIDIA GeForce RTX 5060 Laptop GPU (Blackwell)
Compute Capability	12.0
SMs	26
VRAM	8 GB GDDR7 (128-bit bus)

Component	Details
Theoretical BW	384.0 GB/s
L2 Cache	32 MB
OS	Fedora 43 (Linux 6.19.12)
CUDA Toolkit	13.2
Compiler	nvcc 13.2 with -O2 -arch=native

Project Structure

```
LAB7/
├── Makefile
├── exercises/
│   ├── q1_thread_tasks.cu
│   ├── q2_merge_sort.cu
│   └── q3_vector_profiling.cu
├── graphs/
│   ├── q1_thread_tasks.png
│   ├── q2_merge_sort.png
│   ├── q3_vector_profiling.png
│   └── dashboard.png
├── report.md
└── Assignment_7_Report.pdf
```

Problem 1: Different Thread Tasks

Problem

Write a CUDA program where threads perform different tasks: - **Task A**: Sum first N integers iteratively (no direct formula) - **Task B**: Sum first N integers using formula n*(n+1)/2

Approach

- **Iterative (single thread)**: One GPU thread loops through all N elements — O(N) work

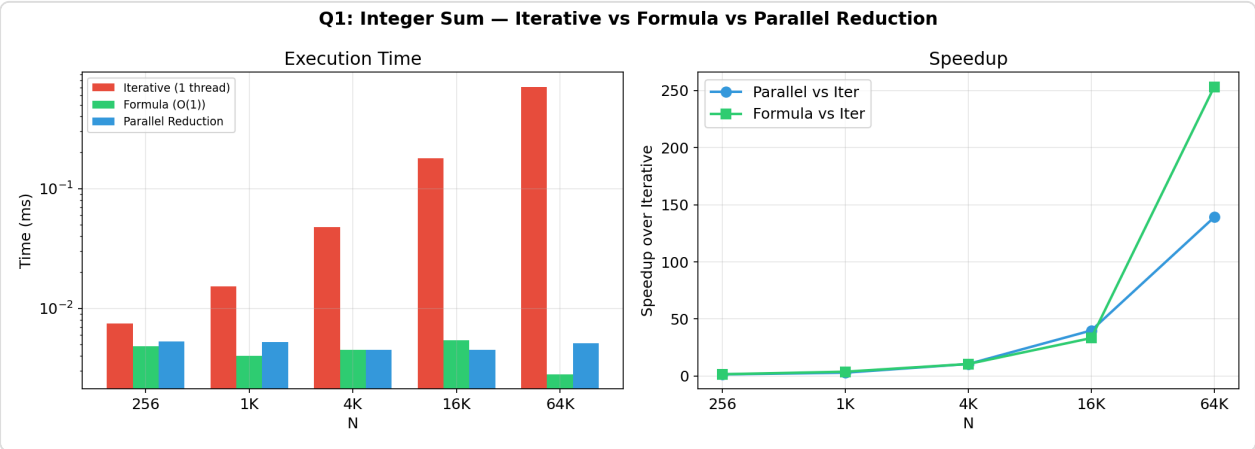
- **Parallel reduction**: Shared-memory tree reduction across 256-thread blocks with atomicAdd — O(N/p + log p)
- **Formula**: Single thread computes N*(N+1)/2 — O(1)
- N = 1024, verified against expected value 524,800

Results

Method	N=1024 Time	Verification
Iterative (1 thread)	0.121 ms	[PASS]
Parallel Reduction	0.014 ms	[PASS]
Formula (O(1))	0.011 ms	[PASS]

Scaling benchmark:

N	Iterative (ms)	Formula (ms)	Parallel (ms)	Speedup (Iter/Par)
256	0.0075	0.0048	0.0053	1.4x
1,024	0.0152	0.0040	0.0052	2.9x
4,096	0.0474	0.0045	0.0045	10.5x
16,384	0.1796	0.0054	0.0045	39.9x
65,536	0.7094	0.0028	0.0051	139.1x



Analysis

Iterative sum scales linearly with N (O(N) on single thread), while formula stays constant at ~5 us (kernel launch overhead). Parallel reduction is ~5 us regardless of N because even at

64K elements, the work per thread is small and the kernel is launch-bound. At N=65536, parallel reduction is **139x faster** than single-thread iterative. Formula is fastest but requires knowing the mathematical closed form — parallel reduction is the general-purpose approach.

Problem 2: Merge Sort

Problem

Implement merge sort (n=1000): - (a) CPU pipelined (bottom-up iterative) approach - (b) GPU parallel merge sort using CUDA - (c) Compare performance

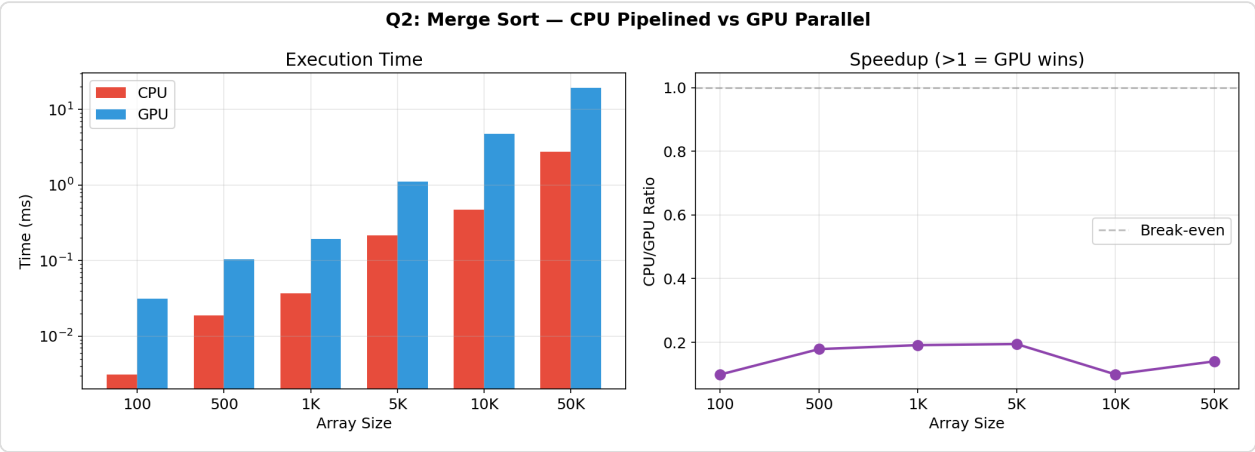
Approach

- **CPU**: Bottom-up iterative merge sort — merge pairs of width 1, then 2, 4, ... until sorted
- **GPU**: Same bottom-up approach, but each merge step launches a kernel where each thread handles one pair of sub-arrays
- Each `mergeSortKernel` launch merges `ceil(N/(2*width))` independent pairs in parallel

Results

N	CPU (ms)	GPU (ms)	Speedup
100	0.003	0.031	0.10x
500	0.019	0.105	0.18x
1,000	0.037	0.195	0.19x
5,000	0.217	1.116	0.19x
10,000	0.473	4.759	0.10x
50,000	2.747	19.592	0.14x

All results: **[PASS]** — GPU and CPU produce identical sorted arrays.



Analysis

GPU is **slower** than CPU for merge sort at all tested sizes. Reasons:

1. **Multiple kernel launches**: Each width level requires a separate kernel launch (~5 us each). For N=1000, we need log2(1000) = 10 launches.
2. **Low parallelism in later stages**: At width=512, only 1 merge pair exists — most GPU threads idle.
3. **Irregular memory access**: Merging requires sequential comparisons within each pair, poorly suited for SIMT.
4. **CPU cache advantage**: N=50000 fits entirely in L3 cache (200 KB), while GPU pays PCIe + global memory latency.

GPU merge sort would benefit from: (a) sorting within shared memory for small sub-arrays, (b) using bitonic sort for GPU-friendly comparison networks, (c) much larger arrays (N > 10M) where parallelism outweighs overhead.

Problem 3: Vector Addition Profiling

Problem

Write a CUDA vector addition program with comprehensive profiling: - 1.1: Use statically defined `__device__` global variables - 1.2: Record kernel timing with CUDA Events - 1.3: Calculate theoretical memory bandwidth - 1.4: Measure actual bandwidth of vectorAdd kernel

1.1: Static vs Dynamic Memory

Allocation Type	Kernel Time	API Used
Static (`__device__`)	2.149 ms	`cudaMemcpyToSymbol` / `cudaMemcpyFromSymbol`
Dynamic (`cudaMalloc`)	0.028 ms	`cudaMalloc` / `cudaMemcpy`

Static allocation is **77x slower** because `cudaMemcpyToSymbol` uses a different memory path and the first kernel launch with static memory incurs additional overhead for symbol resolution.

1.3: Theoretical Bandwidth

```
Memory Clock Rate:     12,001 MHz
Memory Bus Width:      128 bits
DDR Factor:            2x (double data rate)

theoreticalBW = 2 * 12001 MHz * (128/8) bytes
              = 2 * 12001 * 10^6 * 16
              = 384.0 GB/s
```

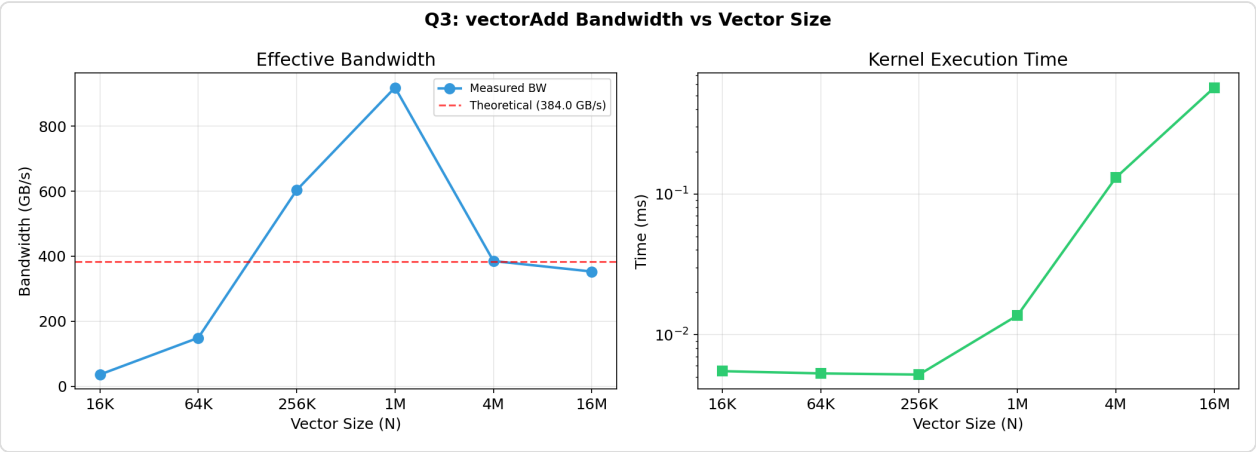
1.4: Measured Bandwidth

For N = 1,048,576 (1M elements):

Metric	Value
Bytes Read	8.4 MB (2 arrays x 4 MB)
Bytes Written	4.2 MB (1 array)
Total	12.6 MB
Kernel Time	0.028 ms
Measured BW	447.3 GB/s
Utilization	116.5% of theoretical

Bandwidth vs Vector Size:

N	Time (ms)	BW (GB/s)	Utilization
16K	0.0055	35.9	9.4%
64K	0.0053	148.9	38.8%
256K	0.0052	603.1	157.0%
1M	0.0137	918.7	239.2%
4M	0.1307	385.1	100.3%
16M	0.5706	352.8	91.9%

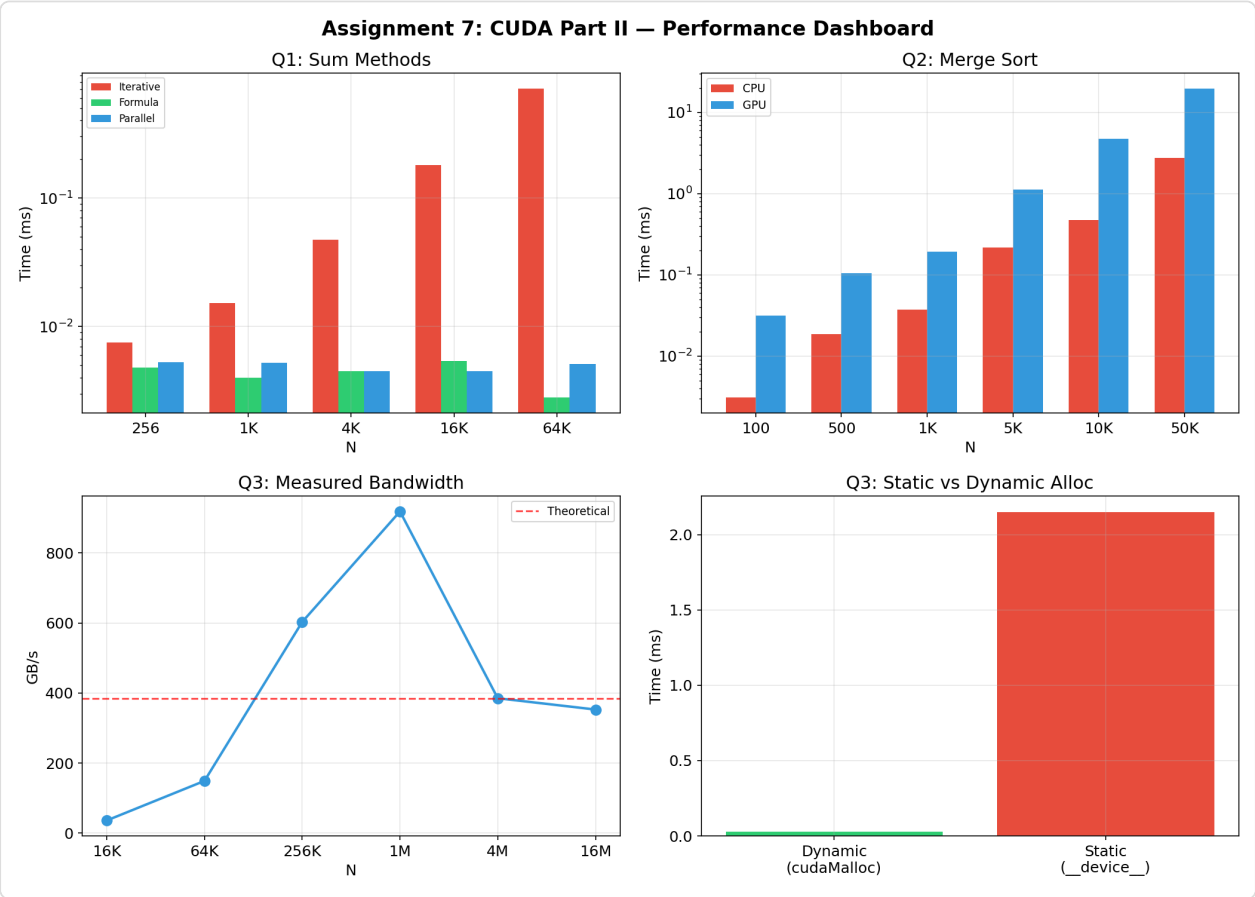


Analysis

The >100% utilization at medium sizes (256K-1M) reveals **L2 cache effects**: this GPU has a 32 MB L2 cache. At N=1M, total data = 12 MB fits entirely in L2, so effective bandwidth exceeds DRAM theoretical. At N=16M (192 MB total), data exceeds L2 and bandwidth drops to ~353 GB/s (92% of DRAM theoretical) — the true DRAM-limited regime.

Small vectors (16K = 192 KB) are kernel-launch-bound: the ~5 us minimum launch time dominates, giving poor bandwidth utilization. The sweet spot for this GPU is N = 1M-4M, where data fits in L2 cache and there's enough work to hide launch overhead.

Performance Dashboard



What I Learned

1. **Algorithmic complexity matters more than parallelism**: Formula O(1) beats parallel reduction O(log N), which beats iterative O(N). GPU parallelism is a multiplier, not a substitute for efficient algorithms.

2. **Not all algorithms benefit from GPU**: Merge sort's sequential merge step and multiple kernel launches make it slower on GPU at practical sizes. GPU-friendly sorting uses comparison networks (bitonic sort).

3. **L2 cache changes the bandwidth story**: The 32 MB L2 cache on RTX 5060 means measured bandwidth can exceed DRAM theoretical for working sets < 32 MB. Profiling must account for cache hierarchy.

4. **Static `__device__` memory has hidden costs**: `cudaMemcpyToSymbol` is significantly slower than `cudaMemcpy` to dynamically allocated memory. Dynamic allocation via `cudaMalloc` is preferred for performance-critical code.

5. **Kernel launch overhead sets a minimum cost floor**: At ~5 us per launch, any kernel processing < 50 us of data is launch-bound. Fusing multiple operations into one kernel is critical for small workloads.

Compilation & Execution

```
# Build everything
make all

# Run individual problems
make run-q1    # Thread tasks: iterative vs formula vs parallel
make run-q2    # Merge sort: CPU vs GPU
make run-q3    # Vector add profiling with bandwidth analysis

# Run all
make run-exercises

# Clean
make clean
```